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Residual Allergen Validation Workbook & Implementation Guide

STEP-BY-STEP GUIDE FOR VALIDATION OF RESIDUAL ALLERGENIC MATERIALS WITHIN RECIRCULATING CHOCOLATE ENROBER SYSTEMS

Workbook Edition – WHO Reference Dose Methodology

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Purpose

This workbook-style guide provides a structured approach for evaluating residual allergenic carryover within recirculating chocolate enrober systems.

The guide is intended to support:

- allergen risk assessments
- validation of allergen control measures
- evaluation of purge or flush effectiveness
- production sequencing decisions
- documentation supporting food safety programs

This guide evaluates allergenic material remaining within recirculating chocolate systems including:

- tempering systems
- chocolate reservoirs
- return chocolate loops
- pumps
- pipes
- bottomers
- curtains
- related recirculating components

This guide does not replace:

- GMPs
- sanitation programs
- allergen labeling requirements
- customer-specific requirements
- regulatory obligations

This guide is intended for use alongside a complete allergen management program.

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Scope and Limitations

This validation evaluates the potential for residual allergenic material to remain within a recirculating chocolate system following production of allergen-containing products.

This validation does NOT:

- guarantee absence of allergenic material
- establish “allergen-free” status
- independently justify removal of advisory labeling
- replace sanitation controls
- replace allergen labeling requirements
- establish a regulatory safe harbor

This validation is only applicable when:

- equipment configuration remains unchanged
- production practices remain substantially similar
- product matrices remain comparable
- allergen profiles remain unchanged
- validated procedures continue to be followed

Facilities should evaluate whether additional customer, export, or regulatory requirements apply.

Minimum Prerequisite Programs Recommended Before Validation

- allergen identification,
- production sequencing,
- sanitation procedures,
- label review,
- product traceability.

Suggested Implementation Timeline

Activity	Estimated Time
Product matrix review	1 day
Worst-case selection	1–2 days
Sample collection	dependent on production
Lab turnaround	1–2 weeks
Final interpretation	1 day

Recommended workflow:

1. Define evaluated enrober system
2. Identify allergens introduced into the chocolate matrix
3. Group representative product matrices
4. Identify representative worst-case products
5. Determine serving size and chocolate contribution
6. Calculate WHO-based thresholds
7. Define sampling locations and timing
8. Collect and submit samples
9. Compare results against calculated thresholds
10. Determine whether additional controls are necessary
11. Establish verification and revalidation activities

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Step 1 – Define the Validation Question

The validation should answer the following question:

Can residual allergenic material carried within the recirculating chocolate matrix remain below the selected WHO reference dose exposure threshold following production of allergen-containing products?

This is not solely a sanitation validation.

This is a matrix carryover validation evaluating allergenic material circulating within the chocolate system itself.

Throughout this workbook Milk Chocolate Covered Oreo-type sandwich cookies will be evaluated as an example.

Facility Documentation

Validation Name/ID: _____

Date Initiated: _____

Responsible Department: _____

Equipment/System Evaluated: _____

Step 2 – Define Equipment Scope

Document the complete chocolate recirculation system being evaluated.

Include:

- enrober line
- tempering system
- return chocolate path
- holding tank/reservoir
- curtain
- bottomer
- pumps
- pipes
- augers
- return trays
- difficult-to-clean components

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Equipment Scope Table

Component	Included in Validation	Notes

Step 3 – Identify Applicable Allergens

Separate allergens into categories.

A. Intrinsic / Always-Present Allergens - These allergens are intentionally present within the chocolate matrix. Examples: Milk, Soy Lecithin

B. Variable Carryover Allergens - These allergens may enter the chocolate matrix from enrobed products. Examples: Peanut, Almond, Walnut, Hazelnut, Wheat, Egg, Sesame

C. Non-Applicable Allergens - Allergens absent from the facility. Examples: Fish, Shellfish

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Applicable Allergen Table

[illegible]

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Step 4 – Build the Product/Allergen Matrix

The goal of this step is not to evaluate every individual SKU.

The goal is to group products with similar operational and allergen carryover characteristics in order to identify representative worst-case product matrices for validation.

Different confectionery matrices may present different carryover behaviors depending on equipment design, production practices, allergen concentration, particulate generation, oil release, product integrity, and run duration. Facilities should evaluate product-specific operational behavior when determining representative worst-case conditions. Products with similar operational behavior may potentially be grouped together when scientifically justified.

Examples may include:

- cookie or wafer products generating dry particulates,
- nut-containing products contributing particulates or oils,
- sticky or smearing centers,
- products prone to breakage during enrobing,
- or long-duration seasonal runs.

Facilities should document the rationale used for grouping products and selecting representative matrices.

Representative Product Group Assessment

Representative Product Group	Example Products	Allergen(s) Introduced into Chocolate System	Operational Carryover Concerns	Relative Carryover Risk
<i>Chocolate Covered Cookies</i>	<i>Oreo-type sandwich cookies</i>	<i>Wheat</i>	<i>Cookie crumbs / particulate shedding</i>	<i>Moderate</i>

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Step 5 – Select Representative Worst-Case Products

Worst-case products are products most likely to contribute allergenic material into the chocolate matrix.

Consider:

- allergen concentration
- run duration
- particulate shedding
- oil release
- product breakdown
- smear potential
- seasonal volume

Facilities may use the following scoring approach:

Risk Factor	Low	Moderate	High
Allergen concentration	Trace/minor	Moderate	High concentration
Run duration	Short	Moderate	Long/continuous
Product breakdown	Stable	Occasional breakage	Frequent crumbling/smear
Oil/fat release	Minimal	Moderate	Significant oil release
Particulate shedding	Minimal	Moderate	Heavy crumbs/fines
Seasonal volume	Limited	Moderate	Extended seasonal run
Product texture	Firm/stable	Semi-soft	Sticky/smearing
Historical findings	None	Minor	Prior complaints/findings

Worst-Case Evaluation Table

<i>Chocolate Covered Sandwich Cookies</i>		
Risk Factor	Evaluation	Rationale
<i>Allergen concentration</i>	<i>Moderate</i>	<i>Wheat present throughout cookie center</i>
<i>Run duration</i>	<i>Low</i>	<i>Short production run</i>
<i>Product breakdown</i>	<i>Moderate</i>	<i>Occasional cookie breakage observed</i>
<i>Oil/fat release</i>	<i>Low</i>	<i>Minimal oil contribution</i>
<i>Particulate shedding</i>	<i>Moderate</i>	<i>Cookie crumbs and fines may enter return chocolate</i>
<i>Product texture</i>	<i>Moderate</i>	<i>Dry particulate-generating matrix</i>
<i>Historical findings</i>	<i>None</i>	<i>No prior allergen complaints</i>

Products with the highest combined risk should generally be selected as representative worst-case matrices.

Facilities should document the rationale for selected representative products.

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Risk Factor	Evaluation	Rationale
Allergen concentration		
Run duration		
Product breakdown		
Oil/fat release		
Particulate shedding		
Seasonal volume		
Product texture		
Historical findings		

Risk Factor	Evaluation	Rationale
Allergen concentration		
Run duration		
Product breakdown		
Oil/fat release		
Particulate shedding		
Seasonal volume		
Product texture		
Historical findings		

Worst-Case Selection Justification:

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Step 6 – Determine Chocolate Contribution

Determine the approximate percentage of the finished serving represented by the chocolate matrix.

Chocolate contribution may include:

- enrobed coating
- bottoming chocolate
- decorative drizzle
- recirculated chocolate inclusions

Use:

- formulation data
- finished weight data
- documented estimates

Use conservative assumptions where variability exists.

Chocolate Contribution Worksheet

Product	Serving Size	Chocolate Weight	Chocolate Fraction
<i>Chocolate Covered Cookies</i>	<i>28 g</i>	<i>11.2 g</i>	<i>40%</i>

Step 7 – Determine WHO Reference Dose Thresholds

WHO reference dose (RfD) thresholds should be used as the basis for exposure calculations.

The calculation converts the WHO reference dose into an allowable ppm concentration within the chocolate matrix.

Lab results are generally expressed as parts per million (ppm). For laboratory reporting purposes, 1 ppm is generally equivalent to 1 mg/kg of sample matrix.

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WHO Reference Dose Table

Allergen	WHO RfD (Reference Dose - mg total protein)
Walnut/Pecan	1.0
Cashew/Pistachio	1.0
Almond	1.0
Peanut	2.0
Egg	2.0
Hazelnut	3.0
Wheat	5.0

Source: [WHO Risk Assessment of Food Allergens Part 2: Review and Establish Threshold Levels for the Priority Allergens \(ISSN 2415-1173\)](#). (Page 94)

Facilities should recognize:
reference doses are risk assessment tools,
regulatory acceptance may vary,
customer requirements may be more restrictive,
and no universally accepted regulatory threshold exists for all allergens.

The selected threshold source and rationale should be documented within the validation.

Threshold Calculation Formula

Allowable ppm in chocolate matrix:

$$\text{Allowable ppm} = \frac{\text{Reference Dose (mg)}}{\text{Serving Size (kg)} \times \text{Chocolate Fraction}}$$

Example: Chocolate Covered Cookies

$$446.4 \text{ Allowable ppm} = \frac{5.0 \text{ Reference Dose (mg)}}{0.028 \text{ Serving Size (kg)} \times 0.40 \text{ Chocolate Fraction}}$$

A laboratory result below 446 ppm would remain below the selected WHO reference dose exposure threshold for this example

Calculation Worksheet

Item: Chocolate Covered Cookies

Serving Size (g): 28g

Serving Size (kg): 28g x 0.001 = 0.028 kg

Chocolate Fraction: 40% or 0.40

WHO RfD (mg): 5.0

Calculated Threshold (ppm): 446.4

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Item: _____

Serving Size (g):

Serving Size (kg):

Chocolate Fraction:

WHO RfD (mg):

Calculated Threshold (ppm):

Item: _____

Serving Size (g):

Serving Size (kg):

Chocolate Fraction:

WHO RfD (mg):

Calculated Threshold (ppm):

Item: _____

Serving Size (g):

Serving Size (kg):

Chocolate Fraction:

WHO RfD (mg):

Calculated Threshold (ppm):

Item: _____

Serving Size (g):

Serving Size (kg):

Chocolate Fraction:

WHO RfD (mg):

Calculated Threshold (ppm):

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Step 8 – Define Sampling Locations

Minimum Sampling Expectations

Sampling should be sufficient to reasonably characterize allergen carryover variability within the evaluated chocolate system.

Recommended minimum sampling:

- minimum 2 sampling locations per evaluated run,
- minimum 3 independent production events for high-risk allergens,
- duplicate samples preferred when feasible,
- retained reserve sample recommended.

High-risk allergens or seasonal worst-case matrices may require additional sampling events to adequately characterize variability.

Facilities should document:

- date,
- product,
- allergen,
- run duration,
- chocolate type,
- production conditions,
- sample location,
- operator,
- and any unusual operational conditions.

Sampling should represent the recirculating chocolate system.

Recommended locations:

- enrober curtain
- return chocolate
- holding reservoir
- post-purge chocolate

Sampling Location Table

Sample Point	Reason for Selection
<i>Enrober curtain</i>	<i>Closest point of chocolate application</i>
<i>Return Chocolate</i>	<i>Evaluates allergen recirculation within return loop</i>

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Evaluating Multiple Chocolate Systems

Separate validations may be required when chocolate systems differ significantly in:

- formulation,
- viscosity,
- fat composition,
- allergen exposure,
- recirculation design,
- shared equipment configuration,
- or production usage.

Examples may include:

- milk chocolate vs dark chocolate,
- compound coating vs real chocolate,
- dedicated vs shared tempering systems.

Facilities should document the rationale when determining whether one validation can reasonably represent multiple chocolate systems.

Step 9 – Define Sampling Timing and Sample Collection Procedures

Chocolate Sample Collection Guidance

Chocolate samples should be collected in a manner minimizing contamination, degradation, or alteration of the matrix.

Facilities should define:

- sample quantity,
- sample container type,
- labeling method,
- cooling requirements,
- storage conditions,
- and shipment conditions.

Recommended practices:

- use clean food-grade sample containers,
- avoid cross-contamination during collection,
- collect representative flowing chocolate rather than surface residue,
- allow chocolate to solidify prior to shipment if required by the laboratory,
- clearly label allergen target and sample location,
- maintain chain-of-custody records.

Facilities should confirm with the selected laboratory:

- acceptable sample quantity,
- shipping temperature,
- required turnaround,
- and matrix compatibility.

Worst-case sampling is typically:

- end of production run
- before significant dilution
- before teardown
- immediately following purge validation

Residual allergenic material may accumulate over time within recirculating chocolate systems.

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Sampling Timing Documentation:

[illegible]

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Step 10 – Laboratory Method Selection

Laboratory Method and Selection Considerations

Facilities should utilize laboratories experienced in allergen analysis of chocolate or high-fat food matrices. The [University of Nebraska-Lincoln Food Allergy Research and Resource Program \(FARRP\)](#) is one example of a laboratory experienced in allergen analysis of chocolate and high-fat food matrices.

Prior to submission, facilities should confirm:

- the laboratory can analyze chocolate matrices,
- the method is validated or suitable for the target allergen,
- reporting units are understood,
- the limit of quantitation supports the calculated threshold,
- and extraction limitations for high-fat products are understood.

Facilities should document:

- laboratory name,
- analytical method,
- analyte basis,
- detection limit,
- quantitation limit,
- and matrix limitations identified by the laboratory.

Laboratory Documentation Table

Sample ID	Laboratory	Matrix	Method	Lower Limit of Quantitation
<i>Milk Chocolate Enrober Curtain [Julian Date of Production]-Sample ID Number</i>	<i>FARRP</i>	<i>Wheat</i>	<i>R-Biopharm Ridascreen® FAST Gliadin</i>	<i>10 ppm</i>

Print and attach ISO 17025 Accreditation Certificate and Analytical reports in lieu of completing table above.

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Step 11 – Collect and Evaluate Results

Compare laboratory results against calculated WHO-based thresholds.

Results Table

[illegible]

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Step 12 – Evaluate Variability and Trends

Evaluate:

- highest observed results
- consistency
- accumulation trends
- operational conditions
- seasonal impacts

Large variability may indicate:

- inconsistent purge effectiveness
- inconsistent production practices
- insufficient controls

Trend Evaluation Notes:

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Step 13 – Define Required Risk Controls

If results exceed calculated thresholds, additional controls may be necessary.

Potential controls:

- validated purge procedures
- dedicated chocolate
- sequencing restrictions
- product hold
- additional sanitation
- revised labeling decisions

Required Corrective Actions

Finding	Required Action	Responsible Person	Completion Date

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Step 14 – Establish Verification Activities

Validation is scientific proof.

Verification confirms the validated process continues functioning as intended.

Verification activities may include:

- production sequencing (allergen changeover matrix)
- purge logs
- sanitation verification
- allergen inspections
- label review
- annual risk review

Verification Records

Verification Activity	Frequency	Responsible Department

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Step 15 – Define Revalidation Triggers

Revalidation should occur when:

- new allergens are introduced
- formulas change
- serving sizes change
- chocolate percentages change
- equipment changes
- purge procedures change
- complaints occur
- customer/regulatory findings occur

Revalidation Trigger Worksheet

Change/Event	Revalidation Required	Notes

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Step 16 – Final Conclusion

The conclusion should avoid overstatement.

Recommended Conclusion Language

Based on the evaluated product matrices, sampling locations, laboratory results, and WHO reference dose calculations, residual allergen carryover within the recirculating chocolate matrix was found to be [below / above] the established threshold for the evaluated allergen(s).

This validation supports the facility's allergen control strategy when used in conjunction with:

- sanitation controls
- production sequencing
- label review
- purge procedures
- routine verification activities

Residual allergen validation is one component of an overall allergen management program. Validation conclusions should be reviewed in conjunction with the facility's overall allergen management program, sanitation capability, labeling decisions, customer requirements, and operational practices.

Facility Conclusion:

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Recommended Conclusion Template

Based on the evaluated product matrices, sampling locations, laboratory results, and WHO reference dose (RfD) calculations, residual allergen carryover within the evaluated recirculating chocolate matrix was determined to be:

- ☐ Below the calculated WHO-based threshold for all evaluated allergen matrices
- ☐ Below the calculated WHO-based threshold for most evaluated allergen matrices with identified exceptions requiring additional controls
- ☐ Above the calculated WHO-based threshold for one or more evaluated allergen matrices

The evaluated enrober system included the following recirculating chocolate components:

Representative worst-case product matrices evaluated included:

Sampling was performed under conditions intended to represent the highest reasonably foreseeable allergen carryover conditions, including:

- ☐ End-of-run sampling
- ☐ Post-purge sampling
- ☐ Extended production duration
- ☐ Seasonal production conditions
- ☐ Other: _____

Laboratory results demonstrated the following:

- ☐ Consistent residual allergen concentrations below calculated thresholds
- ☐ Variable residual allergen concentrations requiring additional evaluation
- ☐ Elevated residual allergen concentrations requiring additional corrective controls

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Where residual allergen concentrations exceeded or approached calculated thresholds, the following additional controls were implemented or recommended:

- ☐ Validated purge/flush procedure
- ☐ Dedicated chocolate or dedicated equipment
- ☐ Revised production sequencing
- ☐ Additional sanitation controls
- ☐ Product hold and evaluation procedures
- ☐ Advisory labeling review
- ☐ Additional verification sampling
- ☐ Other: _____

This validation supports the facility's allergen control strategy when used in conjunction with:

- sanitation programs,
- production sequencing controls,
- allergen labeling review,
- purge procedures,
- employee practices,
- and routine verification activities.

This validation does not:

- guarantee absence of allergenic material,
- establish allergen-free status,
- replace GMPs or sanitation controls,
- or independently justify removal of advisory labeling.

This validation is specific to the evaluated equipment, products, production conditions, serving sizes, and operational assumptions documented within this assessment.

Additional comments or limitations:

Prepared By: _____ Date: _____

Reviewed By: _____ Date: _____

Approved By: _____ Date: _____

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Frequently Asked Questions

Why test chocolate instead of only swabbing equipment?

Because the concern is allergenic material circulating within the recirculating chocolate matrix itself, not solely visible residue on equipment surfaces.

Why sample at the end of the run?

Residual allergenic material may accumulate over time within recirculating chocolate systems and is often highest under end-of-run conditions.

Why not test every product?

Facilities may use representative worst-case products when scientifically and operationally justified.

Products with similar:

- allergen profiles,
- operational behavior,
- particulate generation,
- oil release,
- and production conditions

may potentially be grouped together within the validation rationale.

What makes a product “higher risk” for carryover?

Carryover risk may be influenced by:

- product breakdown,
- particulate shedding,
- oil release,
- smear potential,
- run duration,
- allergen concentration,
- and equipment-specific operational behavior.

Facilities should evaluate actual production conditions when determining representative worst-case products.

Does advisory labeling replace validation?

No. Advisory labeling does not replace reasonable allergen control measures, sanitation controls, or allergen risk assessment activities.

What if results exceed the calculated threshold?

Additional controls may be necessary, including:

- validated purge procedures,
- revised production sequencing,
- additional sanitation,
- dedicated chocolate systems,
- additional sampling,

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- or other corrective actions.

What if results are BLQ or ND?

Facilities should confirm the laboratory detection limits adequately support the calculated WHO-based threshold.

Why use WHO reference doses?

WHO reference doses provide a scientifically recognized framework for contextualizing allergen exposure relative to serving size and chocolate contribution within the finished product.

Can this validation guarantee absence of allergens?

No.

This validation evaluates residual allergen carryover under the documented operating conditions. It does not establish allergen-free status or guarantee absence of allergenic material.

When should the validation be repeated?

Revalidation should be considered when:

- allergens change,
- formulations change,
- equipment changes,
- production practices change,
- purge procedures change,
- or significant complaints/findings occur.

Can Food Safety Bytes provide technical support?

Yes.

Food Safety Bytes may provide technical support related to:

- workbook implementation,
- threshold calculations,
- sampling strategy,
- validation interpretation,
- corrective actions,
- and audit preparation.

For Additional Support

Food Safety Bytes may provide technical support related to workbook implementation, threshold calculations, sampling strategy, validation interpretation, corrective actions, and audit preparation.

[Food Safety Bytes Official Website](#)